User Inputs

Accelerator Design

Design Constraints

Arithmetic Unit Library

Add8

Sub10

Mul11

...

Design Space Construction

Initial Sampling

Synthesis & Execution

Initial Dataset

Models Training
PPA & Accuracy GNNs

Initial models & measured references

Self-Evolving DSE

Search

Candidate Generation (NSGA-III)

Surrogate Prediction

Pareto Ranking

Candidate Selection

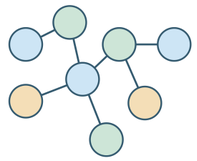
Enough Candidates?

No

Yes

Surrogate Models

Configuration Graph G



PPA GNN

Area / Power Readout

Path-Aware Readout (Latency)

$\hat{A}$

$\hat{P}$

$\hat{L}$

Accuracy GNN

Per-Image SSIM

Mean Accuracy

Local Calibration

Measured References

$\tilde{L}(x) = L(r^*) + \hat{L}(x) - \hat{L}(r^*)$

Predicted Objectives $[\hat{A}, \hat{P}, \tilde{L}, \widehat{Acc}]$

Evaluation

Selected Configurations

Logic Synthesis

Application Execution

Measured PPA & Accuracy

y_t

Budget Reached?

No

Yes

Update

Surrogate Retraining

Reference Update

Updated models & references

Final Search
Fixed Models

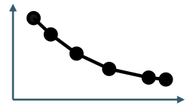
Pareto Front
True PPA & Accuracy

Constraints Met?

No

Yes

Optimized Approximation



Parameter 1

Parameter 2

...